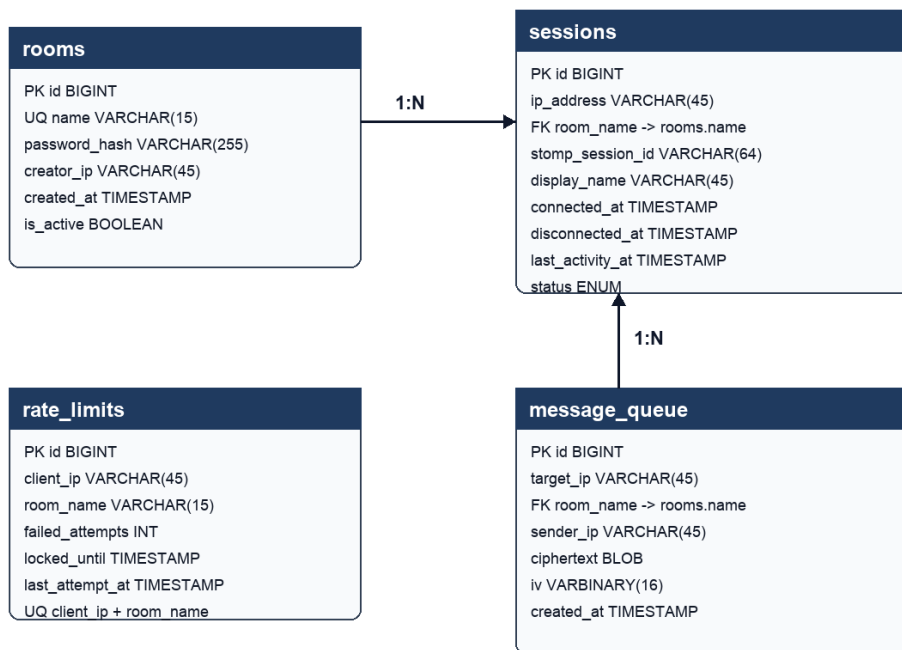


Database Design - Manhattan

ERD - Manhattan Database



Catatan: rate_limits memakai room_name untuk pasangan unik client_ip + room_name, tetapi schema tidak mendefinisikan FK eksplisit.

Entity, Key, dan Relationship

Entity utama: rooms, sessions, message_queue, dan rate_limits. Primary key setiap tabel adalah id. Foreign key eksplisit: sessions.room_name -> rooms.name dan message_queue.room_name -> rooms.name. Relationship utama: rooms 1:N sessions dan rooms 1:N message_queue.

Screenshot Kode

Database Evidence - schema.sql

```
1  -- Manhattan Database Schema
2  -- Tables: rooms, sessions, message_queue, rate_limits
3
4  CREATE TABLE IF NOT EXISTS rooms (
5      id BIGINT AUTO_INCREMENT PRIMARY KEY,
6      name VARCHAR(15) NOT NULL UNIQUE,
7      password_hash VARCHAR(255) NULL,
8      creator_ip VARCHAR(45) NOT NULL,
9      created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
10     is_active BOOLEAN NOT NULL DEFAULT TRUE,
11     INDEX idx_room_name (name)
12 );
13
14 CREATE TABLE IF NOT EXISTS sessions (
15     id BIGINT AUTO_INCREMENT PRIMARY KEY,
16     ip_address VARCHAR(45) NOT NULL,
17     room_name VARCHAR(15) NOT NULL,
18     stomp_session_id VARCHAR(64) NOT NULL,
19     display_name VARCHAR(45) NOT NULL,
20     connected_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
21     disconnected_at TIMESTAMP NULL,
22     last_activity_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
23     status ENUM('ACTIVE', 'INACTIVE') NOT NULL DEFAULT 'ACTIVE',
24     INDEX idx_ip_active (ip_address, status),
25     INDEX idx_room_active (room_name, status),
26     FOREIGN KEY (room_name) REFERENCES rooms(name) ON DELETE CASCADE
27 );
28
29 CREATE TABLE IF NOT EXISTS message_queue (
30     id BIGINT AUTO_INCREMENT PRIMARY KEY,
31     target_ip VARCHAR(45) NOT NULL,
32     room_name VARCHAR(15) NOT NULL,
33     sender_ip VARCHAR(45) NOT NULL,
34     ciphertext BLOB NOT NULL,
35     iv VARBINARY(16) NOT NULL,
36     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
37     INDEX idx_target_room (target_ip, room_name),
38     FOREIGN KEY (room_name) REFERENCES rooms(name) ON DELETE CASCADE
39 );
40
```